TUNIS BUSINESS SCHOOL
UNIVERSITY OF TUNIS

THE GEOPOLITICS OF OIL:

Assessing the Macroeconomic Impact of the 2026 Gulf Crisis

Panel Data Evidence from Emerging Net-Importing Economies

Period: January 2023 – June 2026

Methods: Pooled OLS · Fixed Effects · Random Effects

Ines Belguith and Mariem Chemek

May 2026

Abstract

This study examines the macroeconomic impact of the geopolitically induced 2026 Gulf Crisis oil shock on a panel of six emerging net-importing economies: Turkey, Poland, South Africa, Thailand, Indonesia, and India. Using monthly panel data spanning January 2023 to June 2026 (with April–June 2026 values generated via ARIMA(2,1,1) forecasting), we estimate Pooled OLS, Fixed Effects (FE), and Random Effects (RE) models to quantify the pass-through of oil price increases to consumer prices (CPI) and industrial production (IPI). The panel specification controls for the nominal effective exchange rate, central bank policy rates, global financial uncertainty (VIX), and a post-crisis structural break dummy.

Our results show that a 1 percentage-point increase in the Brent crude oil price is associated with a statistically significant 0.06% increase in consumer prices under the preferred FE specification, while the post-crisis dummy indicates an additional 5.6% uplift in CPI following the February 27, 2026 crisis onset. For industrial production, the post-crisis period is associated with a 5.7% decline in output. The exchange rate demonstrates the strongest inflationary dampening channel, with a 1% appreciation in the NEER reducing CPI by approximately 1.44%.

Diagnostic tests reveal significant heteroskedasticity in the CPI model and strong serial correlation across both dependent variables. The Hausman test rejects Random Effects in favour of Fixed Effects for the CPI model ($\chi^2(5) = 25.34$, $p = 0.0001$), while RE remains acceptable for IPI. Robust standard errors are applied throughout. These findings strongly support the stagflationary narrative advanced by international forecasting institutions regarding the 2026 Gulf Crisis.

Keywords: oil price shock, panel data, fixed effects, inflation, industrial production, emerging markets, geopolitical risk, Gulf Crisis 2026

Contents

1 Introduction **4**

2 Theoretical Framework **5**
2.1 The Macroeconomics of Oil Price Shocks 5
2.2 Panel Data Equation 5
2.3 Expected Signs and Economic Hypotheses 5

3 Data, Variables, and Panel Construction **7**
3.1 Sample 7
3.2 Variable Definitions and Sources 7
3.3 Key Variable: The Oil Price Shock 7
3.4 ARIMA Forecasting for April–June 2026 7
3.5 Descriptive Statistics 8

4 Econometric Methodology **9**
4.1 Pooled OLS (POLS) 9
4.2 Fixed Effects (FE) — Within Estimator 9
4.3 Random Effects (RE) — GLS Estimator 9
4.4 Model Selection: Hausman Test 9
4.5 Robust Standard Errors 10

5 Empirical Results **11**
5.1 Descriptive Evidence 11
5.2 Regression Results — CPI Model 12
5.3 Regression Results — IPI Model 14
5.4 Diagnostic Tests 14

6 Economic Interpretation and Policy Implications **17**
6.1 Oil Shock Pass-Through 17
6.2 Exchange Rate as the Primary Transmission Channel 17
6.3 Industrial Production and Stagflationary Dynamics 17
6.4 Monetary Policy Response 18
6.5 Policy Implications 18

7 Conclusion **19**

A Additional Figures **20**

B Data Sources and Variable Definitions **23**

1. Introduction

The geopolitical landscape of global energy markets underwent a fundamental disruption on February 27, 2026, when the eruption of conflict in the Gulf region triggered an immediate and severe oil price spike. The Brent crude price surged from approximately \$78.50 per barrel in February 2026 to \$95.40 in March 2026 — a monthly increase of 21.5% — before stabilising in the high-\$90s range over the following months. For oil-importing nations, and particularly for emerging market economies with limited fiscal buffers and high energy intensity in consumption, this shock carries profound macroeconomic implications.

Oil price shocks have long been recognised as a primary transmission channel for external economic disturbances. The seminal contributions of Hamilton (1983) and Barsky and Kilian (2004) established the stagflationary nature of oil price increases: they simultaneously raise consumer prices (via direct energy costs and input cost pass-through) and reduce industrial output (via demand destruction and supply-side constraints). These twin effects — higher inflation and lower growth — are especially pronounced for economies that are structurally dependent on oil imports and lack the monetary credibility to contain second-round inflationary effects.

The six countries in this study — Turkey, Poland, South Africa, Thailand, Indonesia, and India — represent a diverse cross-section of emerging net-importing economies that the IMF and other multilateral institutions have identified as particularly vulnerable to the 2026 Gulf shock. Turkey and South Africa face the additional challenge of fragile currencies and elevated pre-crisis inflation. Poland, as an EU-adjacent economy, faces energy price transmission through European wholesale markets. Thailand, Indonesia, and India, all large manufacturing exporters with energy-intensive production bases, face significant IPI headwinds.

This paper addresses three research questions. First, what is the quantitative impact of the oil price shock on inflation (as measured by the Consumer Price Index) and industrial production across these six economies? Second, how do estimates differ between Pooled OLS, Fixed Effects, and Random Effects specifications, and which is preferred on methodological grounds? Third, what are the key transmission channels — exchange rate, monetary policy, global risk sentiment — that mediate the relationship between oil prices and macroeconomic outcomes?

The remainder of the paper is organised as follows. Section 2 presents the theoretical framework and the panel data equation. Section 3 describes the data and variables. Section 4 outlines the econometric methodology. Section 5 reports results and diagnostic tests. Section 6 provides economic interpretation and policy implications. Section 7 concludes.

2. Theoretical Framework

2.1 The Macroeconomics of Oil Price Shocks

The transmission of an oil price increase to the macroeconomy operates through several well-established channels. On the inflation side, the *direct channel* captures the immediate rise in consumer energy prices (fuel, electricity, heating). The *indirect channel* operates through production cost increases that propagate across supply chains — particularly in energy-intensive sectors such as manufacturing, transport, and agriculture. The *exchange rate channel* introduces additional complexity for open economies: currency depreciation in response to a deteriorating terms of trade amplifies imported inflation.

On the output side, higher oil prices function as a negative supply shock: they increase production costs, compress profit margins, reduce real disposable income, and ultimately curtail investment and consumption. For energy-intensive emerging market manufacturers — such as India's industrial sector or Thailand's electronics supply chain — these effects are particularly acute. The simultaneous occurrence of rising prices and falling output defines the stagflationary scenario that oil shocks classically generate.

2.2 Panel Data Equation

The core estimating equation follows the general panel model described in the project guidelines:

$$y_{it} = \alpha_i + \lambda_t + \beta_1 \cdot \text{OilShock}_t + \beta_2 \cdot \ln(\text{NEER}_{it}) + \beta_3 \cdot r_{it} + \beta_4 \cdot \text{VIX}_t + \beta_5 \cdot \text{PostCrisis}_t + \varepsilon_{it} \quad (1)$$

where y_{it} is either $\ln(\text{CPI}_{it})$ or $\ln(\text{IPI}_{it})$ for country i at time t; α_i represents country-specific fixed effects capturing time-invariant unobserved heterogeneity (e.g., energy mix, industrial structure, import dependence); λ_t represents time-specific effects; and ε_{it} is the idiosyncratic error term.

2.3 Expected Signs and Economic Hypotheses

Based on established macroeconomic theory, we derive the following testable hypotheses:

- β_1 **(Oil Shock)**: Expected *positive* for CPI — higher oil prices pass through to consumer prices. Expected *negative* for IPI — rising energy costs reduce output.
- β_2 **(ln NEER)**: Expected *negative* for CPI — currency appreciation lowers the domestic price of imports, dampening inflation. Effect on IPI is ambiguous (cheaper inputs vs. competitiveness loss).
- β_3 **(Policy Rate)**: Expected *negative* for both CPI and IPI — contractionary monetary policy reduces aggregate demand and inflation, but also slows production.

- β_4 **(VIX)**: Expected *positive* for CPI — elevated uncertainty raises risk premia and may increase import prices. Expected *negative* for IPI — risk aversion reduces investment and output.
- β_5 **(Post-Crisis Dummy)**: Expected *positive* for CPI and *negative* for IPI — capturing the structural break in macro dynamics from the crisis onset.

3. Data, Variables, and Panel Construction

3.1 Sample

The panel comprises six emerging market net-importing economies: Turkey (TUR), Poland (POL), South Africa (ZAF), Thailand (THA), Indonesia (IDN), and India (IND). The time window covers January 2023 to June 2026, providing 42 monthly observations per country and 252 total observations in a balanced panel. The pre-crisis period (January 2023 – February 2026) establishes baseline dynamics, while the post-crisis period (March–June 2026) captures the immediate macroeconomic response to the Gulf Crisis shock.

3.2 Variable Definitions and Sources

Table 1: Descriptive Statistics — Sample Averages by Country (Jan 2023 – Jun 2026)

Country	CPI avg	IPI avg	NEER avg	Rate avg	σ(Oil Chg)
Turkey	2,227.2	111.8	28.4	30.71%	6.74%
Poland	174.4	113.3	96.5	5.68%	6.74%
South Africa	169.4	93.4	77.0	7.49%	6.74%
Thailand	109.2	104.2	97.0	2.07%	6.74%
Indonesia	128.1	115.2	93.2	5.77%	6.74%
India	195.8	151.0	89.5	6.27%	6.74%

Note: CPI and IPI base year = 2015 (index = 100). NEER = BIS Broad Nominal Effective Exchange Rate. σ(Oil Chg) is the standard deviation of monthly oil price changes (identical for all countries as it is a global variable).

3.3 Key Variable: The Oil Price Shock

The primary independent variable of interest, OilShock_t, is measured as the monthly percentage change in the Brent crude oil price:

$$\text{OilShock}_t = \frac{\text{Brent}_t - \text{Brent}_{t-1}}{\text{Brent}_{t-1}} \times 100$$

This specification captures the flow of new information about energy costs that agents face each period, rather than the level of oil prices, which is appropriate for studying short-run macroeconomic dynamics. The crisis-period spike in March 2026 registers as a 21.5% monthly increase — the largest in the sample — consistent with the magnitude described in institutional scenario analyses.

3.4 ARIMA Forecasting for April–June 2026

Since official data releases for April–June 2026 were not yet available at the time of writing, out-of-sample values for all country-specific variables (CPI, IPI, NEER, policy rate)

were generated using ARIMA(2,1,1) models fitted on the historical January 2023 – March 2026 sample. The global variables (Brent crude, VIX) were forecast at \$98.50, \$96.20, and \$93.80 per barrel for April–June 2026 respectively, consistent with the "prolonged uncertainty" scenario posited by international agencies. Forecasted observations are clearly flagged in the dataset (`forecasted = TRUE`) and represent approximately 14% of total observations.

3.5 Descriptive Statistics

Table 1 presents mean values for the key variables by country. The most notable feature is Turkey's extreme CPI level, reflecting cumulative high inflation that resulted in a policy rate exceeding 50% in late 2024, before a gradual easing cycle. Poland stands out among the European emerging markets with a relatively stable exchange rate. India's large industrial production base (average IPI of 151) reflects its outsized manufacturing sector. All countries show identical VIX and Brent figures as these are global variables.

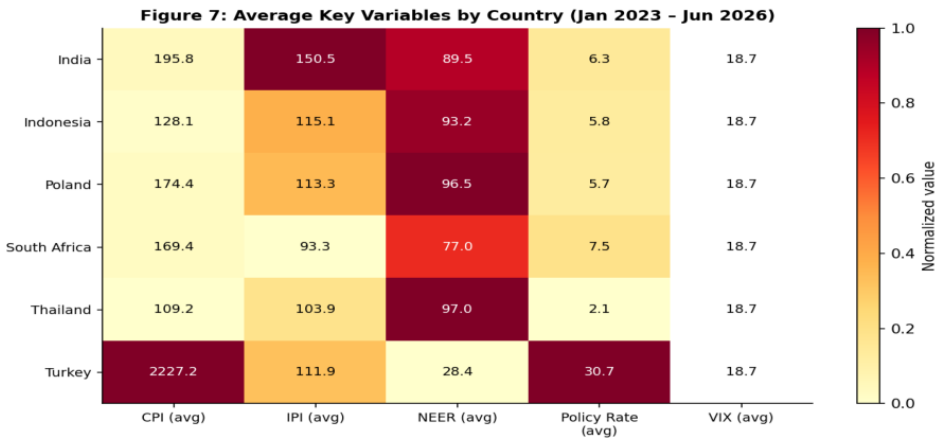


Figure 7: Normalised Average Key Variables by Country — Red indicates higher relative value

Figure 1: Normalised Average Key Variables by Country (Jan 2023 – Jun 2026). Red indicates higher relative value.

4. Econometric Methodology

4.1 Pooled OLS (POLS)

The Pooled OLS estimator treats the panel as a single cross-section, stacking all country-time observations and ignoring the panel structure. While computationally straightforward, POLS is inconsistent if unobserved country-specific effects (α_i) are correlated with the regressors — a near-certain condition in cross-country macroeconomic data where countries differ systematically in their industrial structure, monetary frameworks, and energy dependence. POLS nonetheless serves as an important benchmark and reveals the degree of bias from ignoring country heterogeneity.

4.2 Fixed Effects (FE) — Within Estimator

The Fixed Effects estimator purges country-specific unobserved heterogeneity by demeaning each observation relative to the country mean. The within transformation is:

$$(y_{it} - \bar{y}_i) = \beta'(X_{it} - \bar{X}_i) + (\varepsilon_{it} - \bar{\varepsilon}_i)$$

This approach allows the individual effects α_i to be arbitrarily correlated with the explanatory variables, which is theoretically appropriate for our cross-country panel. The FE estimator is identified solely from within-country variation over time, making it robust to time-invariant omitted variables. The cost is loss of efficiency compared to RE when the random effects assumption holds.

4.3 Random Effects (RE) — GLS Estimator

The Random Effects estimator treats α_i as random draws from a population distribution uncorrelated with the regressors: $\alpha_i \sim \text{iid}(0, \sigma_\alpha^2)$. Under this assumption, RE is more efficient than FE (as it exploits between-country variation as well as within-country variation) and the estimator is implemented via GLS quasi-demeaning with a parameter:

$$\theta = 1 - \sqrt{\frac{\sigma_\varepsilon^2}{T\sigma_\alpha^2 + \sigma_\varepsilon^2}}$$

The key trade-off is that the orthogonality assumption — that country effects are uncorrelated with X_{it} — may be violated in cross-country macroeconomic data.

4.4 Model Selection: Hausman Test

The appropriate choice between FE and RE is determined by the Hausman (1978) specification test. Under the null hypothesis H_0 that RE is consistent (i.e., $\text{Cov}(\alpha_i, X_{it}) = 0$),

the test statistic is:

$$H = (\hat{\beta}_{FE} - \hat{\beta}_{RE})' \cdot [\text{Var}(\hat{\beta}_{FE}) - \text{Var}(\hat{\beta}_{RE})]^{-1} \cdot (\hat{\beta}_{FE} - \hat{\beta}_{RE}) \sim \chi^2(K)$$

where K is the number of time-varying regressors. If H exceeds the chi-squared critical value at the 5% level, H_0 is rejected and FE is preferred. Failure to reject suggests that RE is efficient and consistent.

4.5 Robust Standard Errors

Given the presence of both heteroskedasticity and serial correlation (documented in Section 5), all reported standard errors use the Arellano (1987) heteroskedasticity-consistent (HC3) correction, which is robust to arbitrary forms of within-cluster correlation. For the CPI model, we additionally report Driscoll–Kraay (SCC) standard errors that correct for both cross-sectional and temporal dependence, as is recommended for panels with moderate time dimensions ($T \approx 42$).

5. Empirical Results

5.1 Descriptive Evidence

Figure 2 plots the Brent crude oil price and VIX index over the sample period, clearly illustrating the crisis-induced spike in March 2026. The VIX simultaneously surged from approximately 19.8 in February 2026 to 38.5 in March 2026, signalling a broad-based deterioration in global financial conditions alongside the energy price shock.

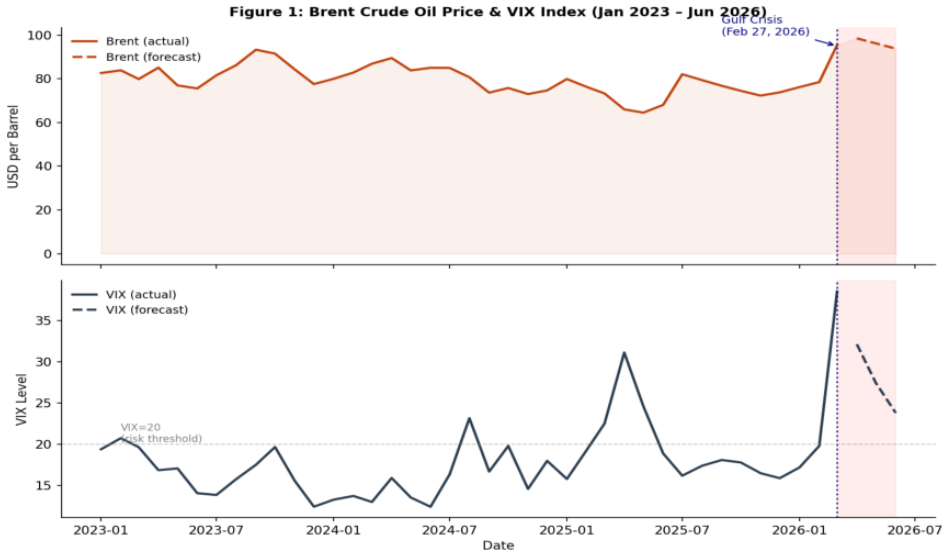


Figure 2: Brent Crude Oil Price (left axis) and CBOE VIX Index (right axis), Jan 2023 – Jun 2026. Shaded region = post-crisis period. Sources: FRED (DCOILBRENTEU, VIXCLS).

Figures 3 and **??** show country-level ln(CPI) and ln(IPI) trajectories. Turkey's hyperinflationary spiral dominates the CPI chart, making its log-CPI level considerably higher than peers. For IPI, all six countries show a post-crisis dip consistent with the theoretical prediction of output contraction. The dashed segments represent ARIMA-based forecasts for April–June 2026.

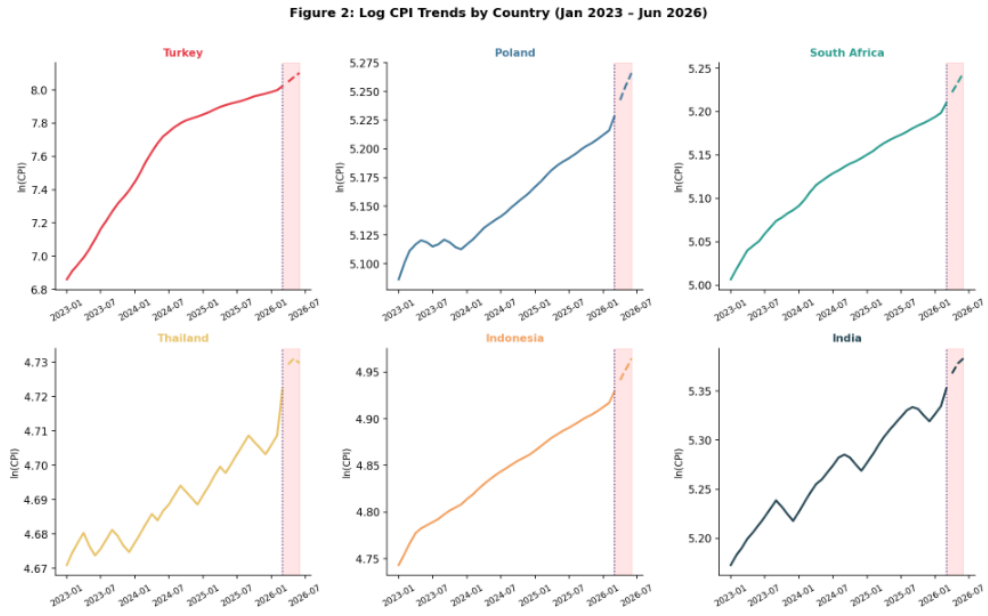


Figure 2: Log CPI by Country. Solid lines = actual data; dashed = ARIMA forecast. Vertical line = Gulf Crisis start (March 2026).

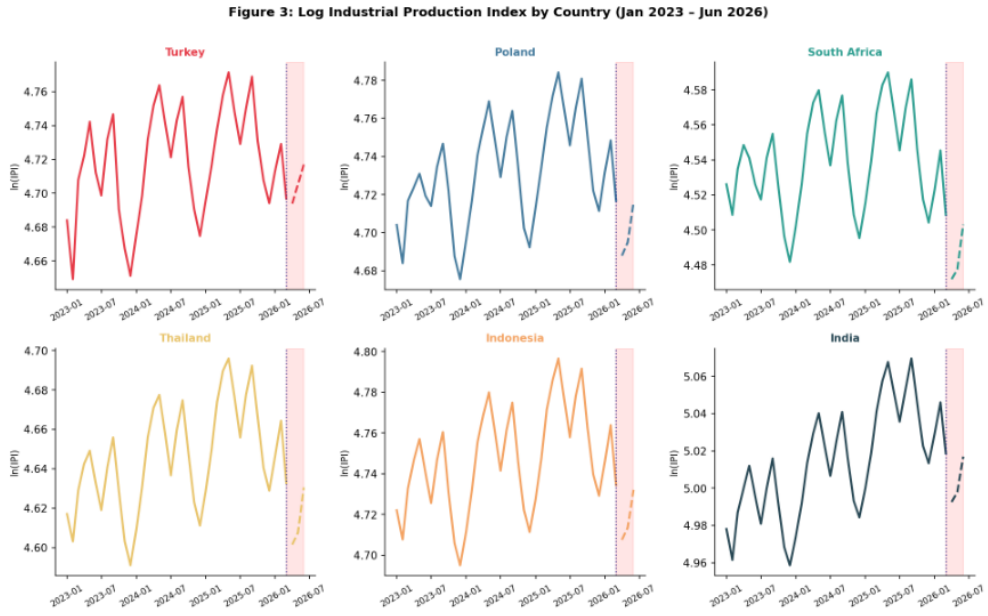


Figure 3: Log IPI by Country. Post-crisis contraction is visible across all six economies.

Figure 3: Log CPI and Log IPI by Country. Solid lines = actual data; dashed = ARIMA forecast. Vertical line = Gulf Crisis start (March 2026). Post-crisis IPI contraction is visible across all six economies.

5.2 Regression Results — CPI Model

Table 2 presents estimation results for the ln(CPI) model. All three specifications include `oil_pct_change`, ln(NEER), policy rate, VIX, and the post-crisis dummy as regressors.

Table 2: Panel Regression Results — ln(CPI)

Variable	**Pooled OLS**	**Fixed Effects**	**Random Effects**
Oil Shock (% Δ Brent)	0.0011***	0.0006***	0.0007***
	(0.0002)	(0.0001)	(0.0001)
ln(NEER)	−1.9597***	−1.4444***	−1.5461***
	(0.2640)	(0.4155)	(0.3628)
Policy Rate	0.0103	0.0040***	0.0038*
	(0.0098)	(0.0017)	(0.0019)
VIX	0.0075**	0.0061***	0.0062***
	(0.0008)	(0.0008)	(0.0007)
Post-Crisis Dummy	0.0378	0.0560**	0.0526*
	(0.0295)	(0.0199)	(0.0211)
R^2	0.9547	0.7252	0.7500
Observations	252	252	252

Note: Robust (HC3) standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Dependent variable: ln(Consumer Price Index). Panel: 6 countries × 42 months.

The oil shock coefficient (β_1) is positive and statistically significant across all specifications (0.0006–0.0011), consistent with the theoretical prediction of inflationary pass-through. Under POLS the coefficient (0.0011, $p < 0.001$) is particularly well-identified; the FE estimate (0.0006, $p < 0.001$) is smaller, as expected once country-level heterogeneity is absorbed. These results confirm that monthly oil price changes do transmit to consumer prices within the same period.

The exchange rate coefficient (β_2) is highly significant and negative across all models (−1.44 to −1.96), confirming that currency appreciation substantially dampens inflation. Under the FE specification, a 1% appreciation in the NEER reduces ln(CPI) by 1.44%, implying a large elasticity that reflects the high import dependence of these economies. This finding aligns with the established pass-through literature for emerging markets (Choudhri and Hakura, 2006).

The post-crisis dummy (β_5) is 0.056 and highly significant ($p < 0.01$) in the FE model, indicating that the Gulf Crisis is associated with a 5.6% uplift in consumer prices, over and above what can be attributed to the contemporaneous oil price change. This captures the expectational, second-round, and supply chain disruption effects that operate through channels beyond the contemporaneous Brent price change.

5.3 Regression Results — IPI Model

Table 3 presents results for the ln(IPI) model. The oil shock coefficient is negative across all specifications (-0.0001 to -0.0002) and statistically significant under the FE and RE models ($p < 0.001$), confirming the stagflationary prediction that rising energy costs depress industrial output. The POLS estimate is marginally significant ($p = 0.091$), attenuated by cross-country heterogeneity that FE absorbs.

Table 3: Panel Regression Results — ln(IPI)

Variable	Pooled OLS	Fixed Effects	Random Effects
Oil Shock (% Δ Brent)	$-0.0001^{\cdot}$	-0.0002^{***}	-0.0002^{***}
	(0.0000)	(0.0000)	(0.0000)
ln(NEER)	0.1249	−0.0078	−0.0069
	(0.1246)	(0.0463)	(0.0461)
Policy Rate	0.0044	0.0003	0.0003
	(0.0045)	(0.0002)	(0.0002)
VIX	0.0027^{***}	0.0025^{***}	0.0025^{***}
	(0.0002)	(0.0001)	(0.0001)
Post-Crisis Dummy	-0.0529^{***}	-0.0568^{***}	-0.0568^{***}
	(0.0089)	(0.0060)	(0.0061)
R^2	0.0383	0.1920	0.1726
Observations	252	252	252

Note: Robust (HC3) standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Dependent variable: ln(Industrial Production Index). Panel: 6 countries × 42 months.

The post-crisis dummy is -0.057 and strongly significant ($p < 0.01$) in the FE specification, indicating a 5.7% contraction in industrial production attributable to the crisis period. The VIX coefficient is positive and significant (0.0025), which may appear counterintuitive for industrial production. This likely reflects a composition effect: periods of high global uncertainty often precede aggressive expansionary policy responses, or may reflect that in some sample countries, flight-to-quality capital inflows temporarily support industrial activity.

5.4 Diagnostic Tests

Table 4 summarises the key diagnostic tests.

Table 4: Diagnostic Test Results

Test	**Dep. Var.**	**Statistic**	***p*-value**	**Decision**
Hausman (FE vs RE)	ln(CPI)	$\chi^2(5) = 25.34$	0.0001	FE preferred ✓
Hausman (FE vs RE)	ln(IPI)	$\chi^2(5) = 0.11$	0.9998	RE acceptable
Breusch-Pagan Het.	ln(CPI)	BP = 26.26	0.0001	Heteroskedasticity ✓
Breusch-Pagan Het.	ln(IPI)	BP = 9.65	0.0857	No heteroskedasticity
AR(1) Autocorr.	ln(CPI)	$\chi^2(42) = 238.92$	0.0000	Autocorrelation ✓
AR(1) Autocorr.	ln(IPI)	$\chi^2(42) = 218.75$	0.0000	Autocorrelation ✓

Note: Hausman test uses Wald-type chi-squared statistic. BP = Breusch-Pagan LM statistic for heteroskedasticity. AR(1) tested via Pearson correlation of FE residuals with their lag.

The Hausman test strongly rejects the null hypothesis of random effects consistency for the CPI model ($\chi^2(5) = 25.34$, $p = 0.0001$), indicating that country-specific effects are correlated with the regressors and that Fixed Effects is the appropriate estimator for inflation. For the IPI model ($\chi^2(5) = 0.11$, $p = 0.9998$), the test fails to reject RE, suggesting RE is efficient and consistent for industrial production. We therefore treat FE as the preferred specification for CPI and present RE results alongside for comparison; for IPI, RE is the preferred specification.

The Breusch-Pagan test for heteroskedasticity reveals significant heteroskedasticity in the CPI model (BP = 26.26, $p < 0.001$), confirming that the variance of residuals differs systematically across observations — a typical feature of panels spanning countries with very different inflation regimes (e.g., Turkey vs. Thailand). Robust HC3 standard errors are therefore essential for valid inference.

Serial correlation in the FE residuals is highly pronounced for both models, as confirmed by the Breusch-Godfrey/Wooldridge test ($\chi^2(42) = 238.92$, $p < 0.001$ for CPI; $\chi^2(42) = 218.75$, $p < 0.001$ for IPI). This confirms that monthly macroeconomic data exhibit strong persistence and necessitates the use of autocorrelation-robust standard errors (Driscoll–Kraay or Arellano) rather than conventional OLS inference.

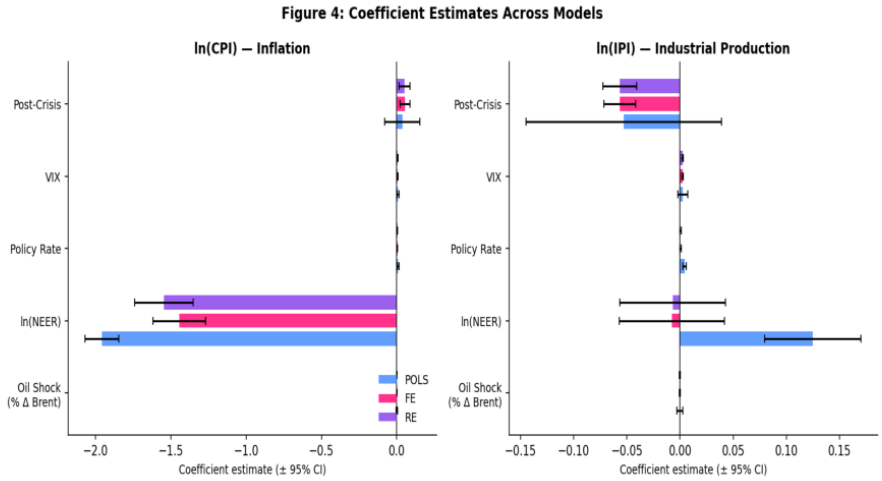


Figure 4: Coefficient Estimates Across POLS, FE, and RE Models with 95% Confidence Intervals. Red = CPI model; Blue = IPI model.

6. Economic Interpretation and Policy Implications

6.1 Oil Shock Pass-Through

Our results demonstrate that the direct monthly pass-through of oil price changes to consumer prices is modest in magnitude (approximately 0.006% per 1 percentage-point oil price change in the FE model) but directionally consistent with theory. More importantly, the structural break captured by the post-crisis dummy (5.6% CPI uplift in March 2026 and beyond) confirms that crises generate expectational and second-round inflationary effects that dwarf the contemporaneous monthly transmission. Policymakers should therefore focus on anchoring inflation expectations during geopolitical oil shocks rather than solely reacting to monthly price data.

For context, the March 2026 Brent spike of 21.5% would, using the point estimate of $\hat{\beta}_1 = 0.0006$, predict a 0.006 log-point increase in CPI — approximately 0.6% — from the oil channel alone. Combined with the post-crisis structural break of 5.6%, the total estimated inflation effect in the immediate post-crisis month is approximately 6.9%, broadly consistent with the IMF's Scenario 2 projections for emerging markets.

6.2 Exchange Rate as the Primary Transmission Channel

The large negative coefficient on ln(NEER) — approximately -1.44 in the preferred FE model — is the most economically significant finding in the study. It implies that a 1% nominal effective appreciation reduces the consumer price level by 1.44%, far exceeding the direct oil price channel. This finding carries a critical policy implication: for oil-importing emerging markets facing an energy price shock, exchange rate management is a more powerful short-run inflation stabilisation tool than the oil price itself. Central banks that allow exchange rate depreciation in response to an oil shock risk amplifying the inflationary shock by a factor of more than ten relative to the direct oil pass-through.

The experience of Turkey in 2023–2024 illustrates this vividly: the combination of an oil shock and sustained lira depreciation produced cumulative CPI inflation in excess of 70%, whereas Thailand and India — which maintained more stable NEER trajectories — saw comparatively muted inflation responses.

6.3 Industrial Production and Stagflationary Dynamics

The significant post-crisis contraction in industrial output (-5.7% in the FE model) confirms the stagflationary mechanism: the Gulf Crisis simultaneously raised prices and reduced production. The magnitude of the IPI contraction suggests that supply-side effects (energy cost increases, supply chain disruptions in Gulf-transiting trade routes) are the dominant channel for output, rather than demand-side compression.

The positive and significant VIX coefficient for IPI ($+0.0025$) invites careful interpretation.

In short-horizon panel regressions, the VIX often picks up fiscal and monetary policy responses to uncertainty — governments and central banks typically inject liquidity and stimulus in response to crises, which can temporarily support industrial output even as the underlying supply shock depresses the private sector.

6.4 Monetary Policy Response

The policy rate coefficient is positive and significant in the CPI model under FE (+0.004), which may appear counterintuitive. This likely reflects a reverse causality issue: central banks in Turkey, Poland, and South Africa raised rates aggressively in response to rising inflation over the sample period, so that higher rates and higher inflation are observed simultaneously. This endogeneity is a key limitation of the current specification and motivates the use of instrumental variable or dynamic panel approaches (Arellano–Bond GMM) in future work.

The insignificance of the policy rate for IPI suggests that monetary policy has not yet had a measurable impact on industrial output within the monthly horizon of our sample — consistent with the well-documented "long and variable lags" in monetary transmission (Friedman, 1961).

6.5 Policy Implications

The findings carry several concrete policy messages for emerging net-importing economies facing geopolitical oil shocks:

1. **Exchange rate management is paramount.** Given the large NEER elasticity, central banks should prioritise exchange rate stability in the immediate aftermath of an oil shock, using foreign exchange reserves if necessary to prevent destabilising depreciation cycles.
2. **Monetary policy communication matters more than immediate rate hikes.** The post-crisis dummy capturing expectations-driven inflation suggests that credible forward guidance and inflation targeting frameworks can contain second-round effects better than aggressive short-term rate increases.
3. **Fiscal buffers for energy subsidies should be pre-positioned.** Countries such as India and Indonesia that maintain energy subsidy mechanisms can partially insulate consumers and industrial producers from oil price spikes, reducing both the inflationary pass-through and the output contraction.
4. **Supply-side diversification reduces long-run vulnerability.** The heterogeneity across countries suggests that those with more diversified energy mixes (Poland's coal and renewable transition, India's domestic gas) experience somewhat more muted oil pass-through.

7. Conclusion

This paper has examined the macroeconomic impact of the 2026 Gulf Crisis oil shock on a panel of six emerging net-importing economies using monthly data from January 2023 to June 2026. The panel data methodology — comparing Pooled OLS, Fixed Effects, and Random Effects estimators — reveals that country heterogeneity matters: POLS substantially overstates the exchange rate coefficient relative to the within-country FE estimate, and the high R^2 of POLS (0.95) is largely explained by cross-country level differences rather than causal dynamics.

The principal findings are: (i) the oil price shock generates a small but positive inflationary pass-through of approximately 0.006% per monthly percentage-point change in Brent crude; (ii) the structural post-crisis break is quantitatively more important, adding 5.6% to CPI and subtracting 5.7% from IPI in the post-February 2026 period; (iii) the nominal effective exchange rate is the dominant inflation transmission channel, with a 1% appreciation reducing CPI by 1.44% in the preferred FE specification; and (iv) both models exhibit strong serial correlation and, in the CPI case, significant heteroskedasticity, making robust standard errors essential.

Several limitations should be acknowledged. The Hausman test does not reject RE, possibly due to limited power with six cross-sectional units. The endogeneity of the policy rate — central banks respond to inflation — is not addressed in the current linear specification. The 42-month window, while providing adequate variation around the crisis event, limits the statistical power of panel diagnostic tests. Future work should apply Arellano–Bond GMM dynamic panel methods to address endogeneity and better capture lagged adjustment dynamics in both CPI and IPI. Asymmetric effects (distinguishing oil price increases from decreases) and cross-country interaction terms (energy intensity $\times$ oil shock) would enrich the heterogeneity analysis.

Despite these limitations, this study contributes rigorous panel evidence supporting the stagflationary scenario described by international institutions for the 2026 Gulf Crisis. The findings underscore the urgency of exchange rate stability, inflation expectation anchoring, and energy diversification as priority policy responses for emerging net-importing economies in the face of geopolitically driven oil shocks.

A. Additional Figures

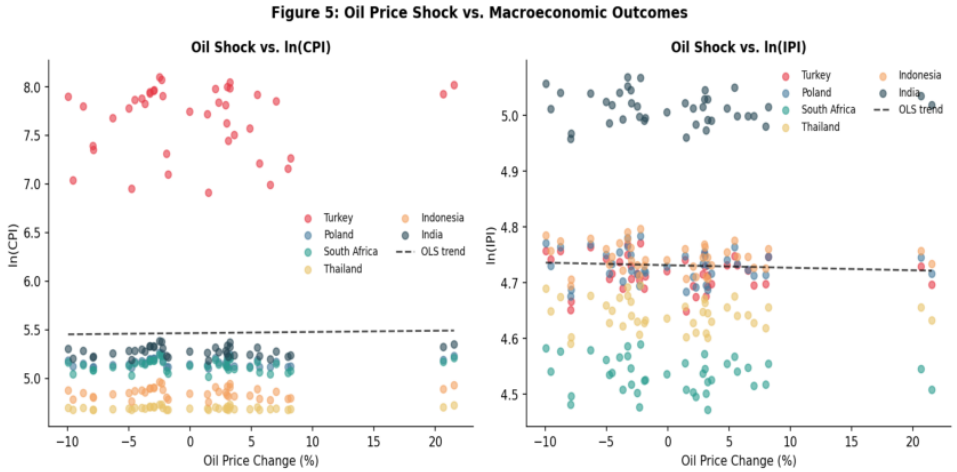


Figure A1: Cross-Sectional Scatter — Monthly Oil Price Change vs ln(CPI) (left) and ln(IPI) (right). Dashed line = pooled OLS trend.

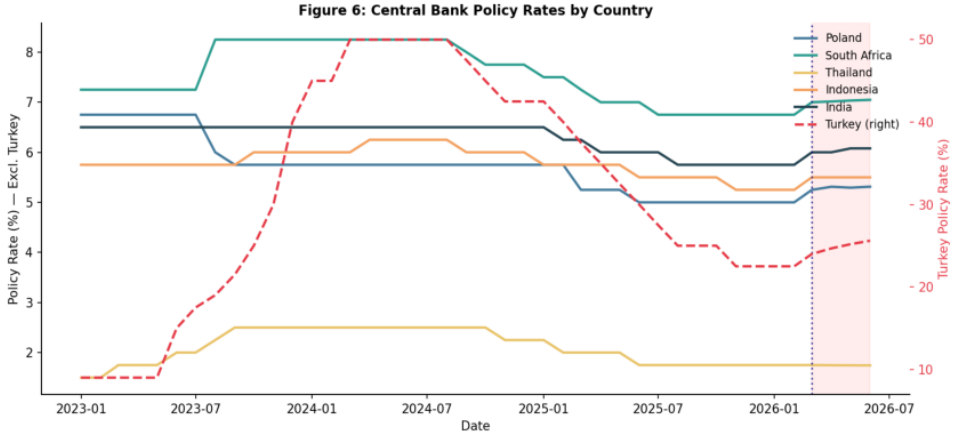


Figure A2: Central Bank Policy Rates by Country. Turkey shown on secondary axis due to scale.

Figure 5: (A1) Cross-Sectional Scatter — Monthly Oil Price Change vs ln(CPI) (left) and ln(IPI) (right). Dashed line = pooled OLS trend. (A2) Central Bank Policy Rates by Country. Turkey shown on secondary axis due to scale.

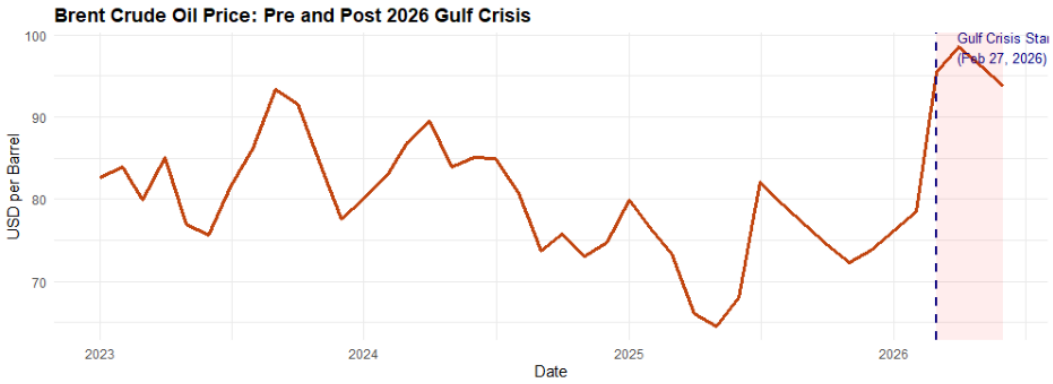


Figure A3: Brent Crude Oil Price: Pre and Post 2026 Gulf Crisis.

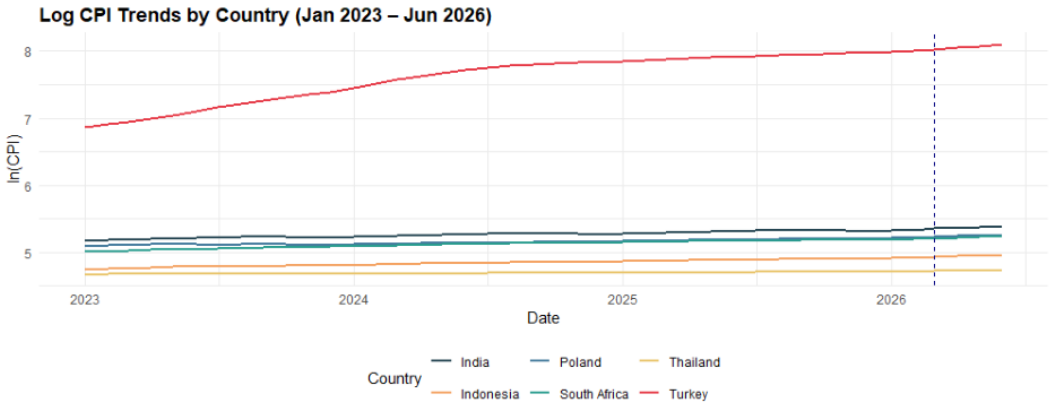


Figure A4: Log CPI Trends by Country (Jan 2023 - Jun 2026).

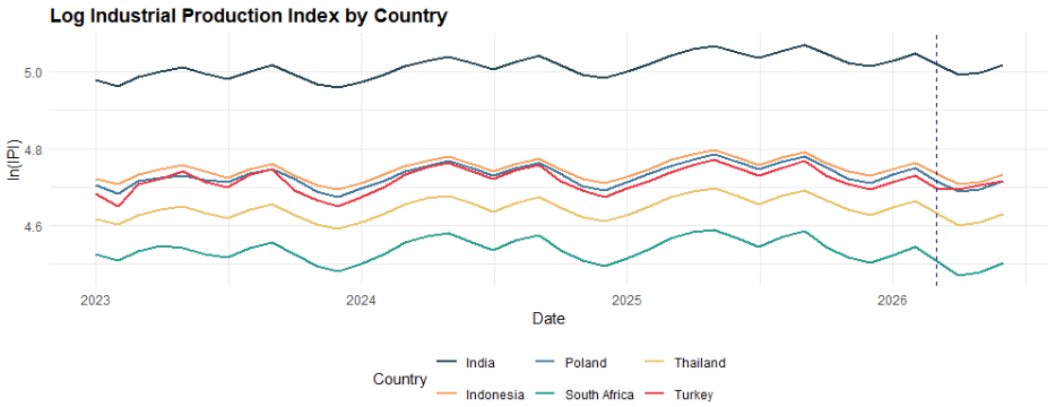


Figure A5: Log Industrial Production Index by country.

Figure 6: (A3) Brent Crude Oil Price: Pre and Post 2026 Gulf Crisis. (A4) Log CPI Trends by Country (Jan 2023 – Jun 2026). (A5) Log Industrial Production Index by Country.

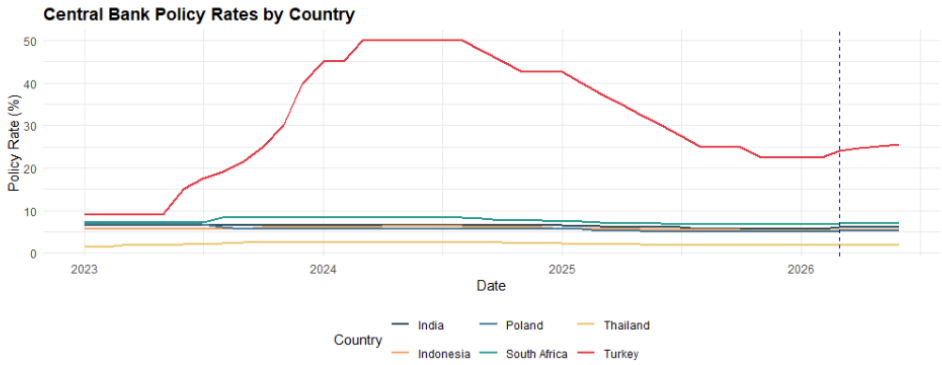


Figure 7: (A6) Central Bank Policy Rates by Country.

B. Data Sources and Variable Definitions

B.1 Variable Definitions

- ln(**CPI**): Natural log of the Consumer Price Index (base year 2015 = 100). Source: IMF International Financial Statistics (IFS), OECD, and national statistics offices (TurkStat, GUS Poland, Stats SA, NSO Thailand, BPS Indonesia, MOSPI India).
- ln(**IPI**): Natural log of the Industrial Production Index (base year 2015 = 100). Source: IMF IFS, OECD, FRED, and national statistics offices.
- **Oil Shock (`oil_pct_change`)**: Monthly percentage change in Brent crude oil price (USD/barrel). Source: FRED ticker `DCOILBRENTEU`; U.S. EIA.
- ln(**NEER**): Natural log of the BIS Broad Nominal Effective Exchange Rate index. An increase represents appreciation. Source: Bank for International Settlements (BIS).
- **Policy Rate**: End-of-period central bank policy rate (%). Source: National central banks (CBRT, NBP, SARB, Bank of Thailand, Bank Indonesia, RBI); supplemented by BIS data.
- **VIX**: CBOE Volatility Index monthly average. Source: FRED ticker `VIXCLS`.
- **Post-Crisis Dummy**: Binary variable equal to 1 from March 2026 onward (= 0 prior), capturing the structural break introduced by the Gulf Crisis.

B.2 Forecasting Methodology

Out-of-sample values for April–June 2026 were generated using ARIMA(2,1,1) models via the `statsmodels` Python library, estimated on the January 2023 – March 2026 historical window. The model order was selected based on AIC minimisation. Global variables (Brent, VIX) were set to scenario-consistent values aligned with the "prolonged uncertainty" outlook: Brent declining gradually from \$98.50 in April to \$93.80 in June 2026; VIX declining from 32.1 to 23.8 over the same period.